CADT IDT

W1 PRACTICE

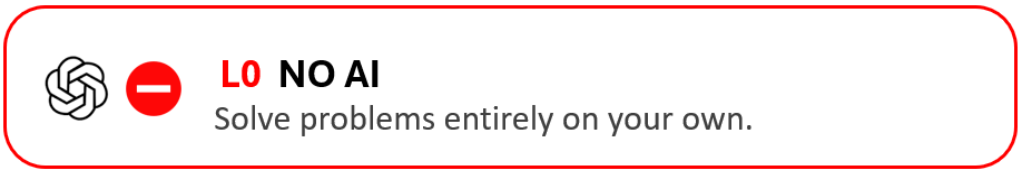
HOMEWORK

⚠ *What is **mandatory** , what is **optional** ?*

- Exercise 1 is **mandatory**
- Exercises 2 and 3 are **optional** is you want to **practice more** or to **go further**

*What do you need to **be able to do** for next week?*

- Identify the **type** of variable
- Manipulate **nullable** and **non-nullable** variables.
- Differentiate between **final** and **const**.
- Manipulate **strings, lists, and maps.**
- Use **loops** and **conditions** to control flow.
- Define and call functions with **positional** and **named arguments**, understand **arrow syntax**



Are you lost?

Read the following documentation to be ready for this practice:

- ✓ Variables
- ✓ Null Safety
- ✓ Built-in types
- ✓ Lists
- ✓ Loops
- ✓ Conditions
- ✓ Functions

EX 1 – Find students who passed {MANDATORY}

Write a Dart program that takes a map of students and their scores and **returns the names of all students who passed**.

Given

- You are given a Map<String, int> where:
  - The key is the student's **name**.
  - The value is their exam **score**.

- A student **passes** if their score is **50 or higher**.

INPUT

```
{
  "Dara": 78,
  "Sokha": 45,
  "Lina": 62,
  "Vanna": 39,
  "Mony": 85
}
```

OUTPUT

```
 [Dara, Lina, Mony]
```

Requirements

- Write a function:

```
List<String> getPassedStudents(Map<String, int> scores)
```

```
List<String> getPassedStudents(Map<String, int> scores) {
 List<String> passedStudents = [];
 scores.forEach((name, score) {
  if (score >= 50) {
   passedStudents.add(name);
  }
 });
 return passedStudents;
}
void main() {
 final scores = {
  "Dara": 78,
  "Sokha": 45,
  "Lina": 62,
  "Vanna": 39,
  "Mony": 85,
 };
```

```
  print(getPassedStudents(scores));
}

// or V2 using arrow
List<String> getPassedStudents(Map<String, int> scores) => scores.entries
    .where((entry) => entry.value >= 50)
    .map((entry) => entry.key)
    .toList();

void main() {
  final scores = {
    "Dara": 78,
    "Sokha": 45,
    "Lina": 62,
    "Vanna": 39,
    "Mony": 85,
  };

  print(getPassedStudents(scores)); // Output: [Dara, Lina, Mony]
}
```

*EX 2 – Compute **courses averages*** *{OPTIONAL}*

Write a Dart program that takes a list of students and their scores for different courses, then **displays the average score <u>for each course</u>**.

Given

Each student is represented by a Map<String, int> containing:

- The student's name.
- Their score for each course.

INPUT

```
final students = [
  {
    "name": "Dara",
    "Dart": 70,
    "Flutter": 80,
    "Database": 60,
  },
  {
    "name": "Sokha",
    "Dart": 50,
    "Flutter": 90,
    "Database": 70,
  },
  {
    "name": "Lina",
    "Dart": 80,
    "Flutter": 70,
    "Database": 90,
  },
];
```

OUTPUT

```
Dart: 66.7
Flutter: 80.0
Database: 73.3
```

Requirements

Divide this work into 2 steps !!

1. Write a function to **compute an average** from a list

```
double computeAverage(List<int> scores)
```

double computeAverage(*List*<*int*> *scores*) {
if (*scores*.isEmpty) return 0;
return *scores*.*reduce*((*a*, *b*) => *a* + *b*) / *scores*.length;
}

2. Process each course

In main: we have created a list containing the course names.
For each course:

- **Create a List<int>** containing the scores of **all students** for that **course only**.
- Pass this list to **computeAverage**().
- Display the course **name** and its **average**.

```
void main() {
  final students = [
    {
      "name": "Dara",
      "Dart": 70,
      "Flutter": 80,
      "Database": 60,
    },
    {
      "name": "Sokha",
      "Dart": 50,
      "Flutter": 90,
      "Database": 70,
    },
  ];
  final courses = ["Dart", "Flutter", "Database"];

  // TODO: Implement computeAverage()
  // TODO: For each course:
  // 1. Create a list of scores
  // 2. Compute the average
  // 3. Display the course name and average
  for (String course in courses) {
      List<int> courseScore= [];

      for (var student in students) {
      if (student.containsKey(course)) {
```

```
      courseScore.add(student[course] as int);
      }
      }
      double average = computeAverage(courseScore);
      print('$course Average: ${average.toStringAsFixed(1)}');
  }
```